

6.course-oss部署

OBE Platform 项目部署文档

运行环境：Ubuntu 22

技术栈：Java 17 + Spring Boot 3 + Vue 3 + MySQL 8.0 + Nginx

一、前置准备

1.1 环境依赖

组件	传统部署	Docker 部署
JDK 17	✓	镜像内置
MySQL 8.0	✓	docker-compose
Nginx	✓	docker-compose
Maven 3.9+	✓（打包）	✓（打包）
Node.js 18+	✓（打包）	✓（打包）
Docker + Compose	—	✓

1.2 打包（两种方式通用）

```
1  # 后端打包
2  cd backend
3  mvn clean package -DskipTests
4  # 产物: backend/target/obe-platform-1.0.0.jar
5
6  # 前端打包
7  cd frontend
8  npm install
9  npm run build
10 # 产物: frontend/dist/
11
12 # 数据库初始化
13 mysql -u root -p < docs/database/init.sql
```

默认管理员: `admin` / `123456`

前端端口: 5173

后端端口: 8080

数据库端口: 3308 (docker映射: 3308: 3306)

账号所在位置: docs\deploy\系统账号.txt

二、方式一：传统部署

2.1 目录结构

```

1  /works/course-oss/
2  |— backend/
3  |   └─ obe-platform-1.0.0.jar      # Maven 打包产物
4  |— frontend/
5  |   └─ dist/                      # npm run build 产物
6  |— nginx/
7  |   └─ logs/                      # Nginx 日志
8  |— logs/
9  |   └─ backend/                  # 后端日志 (application-prod.yml 配置)
10 └─ docs/
11     └─ database/
12         └─ init.sql              # 数据库初始化脚本

```

2.2 systemd 服务

创建 `/etc/systemd/system/obe-platform.service` :

```

1  [Unit]
2  Description=OBE Platform Backend Service
3  After=network.target mysql.service
4
5  [Service]
6  Type=simple
7  User=root
8  WorkingDirectory=/works/course-oss/backend
9  ExecStart=/usr/bin/java -jar /works/course-oss/backend/obe-platform-1.0.0.jar
10 Restart=on-failure
11 RestartSec=10
12
13 [Install]
14 WantedBy=multi-user.target

```

`sudo systemctl daemon-reload`

日志路径已在 `application-prod.yml` 中配置 (`logging.file.name`) , service 文件无需重复声明。

2.3 后端管理

```
1  sudo systemctl start obe-platform      # 启动
2  sudo systemctl stop obe-platform       # 停止
3  sudo systemctl restart obe-platform    # 重启
4  sudo systemctl status obe-platform     # 查看状态
5  sudo systemctl enable obe-platform     # 开机自启
6
7  # 查看日志
8  tail -f /works/course-oss/logs/backend/backend.log
```

2.4 Nginx 配置

直接编辑 `/etc/nginx/nginx.conf` :

```
1  user  www-data;
2  worker_processes  auto;
3
4  error_log  /works/course-oss/nginx/logs/error.log warn;
5  pid        /var/run/nginx.pid;
6
7  events {
8      worker_connections  1024;
9  }
10
11 http {
12     include        /etc/nginx/mime.types;
13     default_type    application/octet-stream;
14
15     log_format  main  '$remote_addr - $remote_user [$time_local] "$request"
16         ,
17         '$status $body_bytes_sent "$http_referer" '
18         '"$http_user_agent" "$http_x_forwarded_for"';
19
20     access_log  /works/course-oss/nginx/logs/access.log  main;
21
22     sendfile    on;
23     tcp_nopush  on;
24     keepalive_timeout  65;
25     gzip         on;
26     gzip_min_length  1k;
27     gzip_types    text/plain application/javascript text/css application/
28     json application/xml;
29
30     server {
31         listen      80;
32         server_name  localhost;
33
34         location / {
35             root      /works/course-oss/frontend/dist;
36             index      index.html;
37             try_files  $uri $uri/ /index.html;
38         }
39
40         location /api/ {
41             proxy_pass http://127.0.0.1:8080;
42             proxy_set_header Host $host;
43             proxy_set_header X-Real-IP $remote_addr;
44             proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
45             proxy_set_header X-Forwarded-Proto $scheme;
```

```
44         client_max_body_size 50m;
45     }
46 }
47 }
```

`sudo nginx -t && sudo systemctl reload nginx`

2.5 启动顺序

```
1  # 1. MySQL
2  sudo systemctl start mysql
3
4  # 2. 初始化数据库 (仅首次)
5  mysql -u root -p < /works/course-oss/docs/database/init.sql
6
7  # 3. 后端
8  sudo systemctl start obe-platform
9
10 # 4. Nginx
11 sudo systemctl start nginx
```

三、方式二：Docker 部署

前端端口：5173

后端端口：8080

数据库端口：3308 (docker映射：3308: 3306)

账号所在位置：docs\deploy\系统账号.txt

3.1 目录结构

```

1 /works/course-oss/
2 |— backend/
3 |   |— Dockerfile          # 后端镜像构建文件
4 |   |— obe-platform-1.0.0.jar # Maven 打包产物
5 |— frontend/              # npm run build 产物

6 |— nginx/
7 |   |— nginx.conf          # 容器内 Nginx 配置
8 |   |— logs/              # Nginx 日志（挂载到宿主机）
9 |— mysql/
10 |   |— data/              # MySQL 数据持久化目录
11 |   |— logs/
12 |       |— backend/      # 后端日志（挂载到宿主机）
13 |— docs/
14 |   |— database/
15 |       |— init.sql      # 数据库初始化脚本
16 |— docker-compose.yml    # 编排文件
17 |— .env                  # 环境变量（可选）

```

● 实验室工位-centos7.9

● 47.96.66.16

←

→

/works/course-oss

名称	大小	类型	修改时间	属性	所有者
..					
backend		文件夹	2026-05-24, 下午 10:20	drwxr-xr-x	root
frontend		文件夹	2026-05-24, 下午 10:20	drwxr-xr-x	root
logs		文件夹	2026-05-24, 下午 10:19	drwxr-xr-x	root
mysql		文件夹	2026-05-24, 下午 10:19	drwxr-xr-x	root
nginx		文件夹	2026-05-24, 下午 10:21	drwxr-xr-x	root
! docker-compose.yml	1KB	Yaml 源文...	2026-05-24, 下午 10:21	-rw-r--r--	root

3.2 Docker 镜像

镜像	用途
openjdk:17.0.2	后端运行环境
nginx:latest	前端 Web 服务器 + 反向代理
mysql:8.0.26	数据库

3.3 Dockerfile

/works/course-oss/backend/Dockerfile:

▼

Dockerfile

```
1  # 基础镜像
2  FROM openjdk:17.0.2
3  # 设定时区
4  ENV TZ=Asia/Shanghai
5  RUN ln -snf /usr/share/zoneinfo/$TZ /etc/localtime && echo $TZ > /etc/timezone
6  # 拷贝jar包
7  COPY obe-platform-1.0.0.jar /app.jar
8  # 入口
9  ENTRYPOINT ["java", "-jar", "/app.jar"]
```

3.4 nginx.conf（容器内使用）

/works/course-oss/nginx/nginx.conf:


```
1  user  root;
2  worker_processes  auto;
3
4  error_log  /var/log/nginx/error.log warn;
5  pid        /var/run/nginx.pid;
6
7  events {
8      worker_connections  1024;
9  }
10
11 http {
12     include        /etc/nginx/mime.types;
13     default_type    application/octet-stream;
14
15     log_format  main  '$remote_addr - $remote_user [$time_local] "$request"
16                       '$status $body_bytes_sent "$http_referer" '
17                       '"$http_user_agent" "$http_x_forwarded_for"';
18
19     access_log  /var/log/nginx/access.log  main;
20
21     sendfile    on;
22     tcp_nopush  on;
23     keepalive_timeout  65;
24     gzip        on;
25     gzip_min_length  1k;
26     gzip_types   text/plain application/javascript text/css application/
27     json application/xml;
28
29     server {
30         listen      5173;
31         server_name  47.96.66.16;
32
33         location / {
34             root      /usr/share/nginx/html/frontend;
35             index      index.html;
36             try_files  $uri $uri/ /index.html;
37             add_header Cache-Control "no-cache, no-store, must-revalidate"; #
38             避免缓存问题
39         }
40
41         location /api/ {
42             proxy_pass http://obe-backend:8080;
43             proxy_set_header Host $host;
44             proxy_set_header X-Real-IP $remote_addr;
```

```
43     proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
44     proxy_set_header X-Forwarded-Proto $scheme;
45     client_max_body_size 50m;
46 }
47 }
48 }
```

3.5 docker-compose.yml

/works/course-oss/docker-compose.yml:

```

1  services:
2    obe-mysql:
3      image: mysql:8.0.26
4      container_name: obe-mysql
5      environment:
6        TZ: Asia/Shanghai
7        MYSQL_ROOT_PASSWORD: Qzb123456*
8      ports:
9        - "3308:3306"
10     volumes:
11       - "/works/course-oss/mysql/data:/var/lib/mysql"
12     networks:
13       - obe-net
14     restart: unless-stopped # 核心: 配置重启策略
15  obe-backend:
16    build:
17      context: ./backend
18      dockerfile: Dockerfile
19    container_name: obe-backend
20    volumes:
21      - "/works/course-oss/logs/backend:/app/logs" # 挂载后端日志 (可选)
22    depends_on:
23      - obe-mysql
24    ports:
25      - "8080:8080"
26    networks:
27      - obe-net
28    restart: unless-stopped # 核心: 配置重启策略
29  obe-nginx:
30    image: nginx
31    container_name: obe-nginx
32    ports:
33      - "5173:5173"
34    volumes:
35      - "/works/course-oss/nginx/nginx.conf:/etc/nginx/nginx.conf"
36      - "/works/course-oss/nginx/logs:/var/log/nginx"
37      - "/works/course-oss/frontend:/usr/share/nginx/html/frontend" # 挂
38    depends_on:
39      - obe-backend
40    networks:
41      - obe-net
42    restart: unless-stopped # 核心: 配置重启策略
43
44  networks:

```

载Vue静态文件

45
46

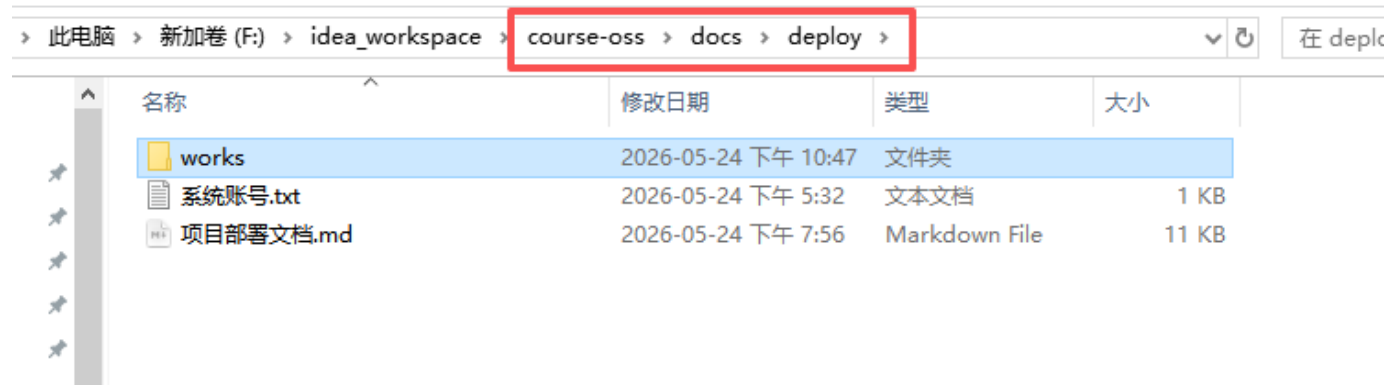
```
obe-net:  
driver: bridge
```

3.6 宿主机与容器目录映射

宿主机路径	容器路径	用途
/works/course-oss/mysql/data	/var/lib/mysql	MySQL 数据持久化
/works/course-oss/docs/database/init.sql	/docker-entrypoint-initdb.d/init.sql	首次启动自动建库建表
/works/course-oss/logs/backend	/app/logs	后端日志
/works/course-oss/nginx/nginx.conf	/etc/nginx/nginx.conf	Nginx 配置（直接挂载）
/works/course-oss/nginx/logs	/var/log/nginx	Nginx 日志

3.7部署

部署所需资源放在~docs\deploy\ 目录下：



将works文件夹放到服务器根目录：

实验室工位-centos7.9						
名称	大小	类型	修改时间	属性	所有者	
bin		文件夹	2026-03-08, 下午 6:20	lr-xr-xr-x	0	
boot		文件夹	2026-03-09, 上午 1:47	dr-xr-xr-x	root	
dev		文件夹	2026-05-24, 下午 10:10	drwxr-xr-x	root	
etc		文件夹	2026-05-24, 下午 10:10	drwxr-xr-x	root	
home		文件夹	2026-03-09, 上午 1:46	drwxr-xr-x	root	
lib		文件夹	2026-03-09, 上午 1:43	lr-xr-xr-x	0	
lib64		文件夹	2026-03-08, 下午 6:20	lr-xr-xr-x	0	
media		文件夹	2018-04-11, 下午 12:59	drwxr-xr-x	root	
mnt		文件夹	2018-04-11, 下午 12:59	drwxr-xr-x	root	
opt		文件夹	2026-03-08, 下午 6:23	drwxr-xr-x	root	
proc		文件夹	2026-05-24, 下午 10:10	dr-xr-xr-x	root	
root		文件夹	2026-05-24, 下午 10:14	dr-xr-x---	root	
run		文件夹	2026-05-24, 下午 10:24	drwxr-xr-x	root	
sbin		文件夹	2026-03-09, 上午 1:44	lr-xr-xr-x	0	
srv		文件夹	2018-04-11, 下午 12:59	drwxr-xr-x	root	
sys		文件夹	2026-05-24, 下午 10:10	dr-xr-xr-x	root	
tmp		文件夹	2026-05-24, 下午 10:22	drwxrwxr...	root	
usr		文件夹	2026-03-09, 上午 1:39	drwxr-xr-x	root	
var		文件夹	2026-03-09, 上午 1:48	drwxr-xr-x	root	
work		文件夹	2026-03-08, 下午 6:54	drwxr-xr-x	root	
works		文件夹	2026-05-24, 下午 10:21	drwxr-xr-x	root	

```

1  # 在 /works/course-oss/ 目录下执行
2  cd /works/course-oss/
3
4  # 构建并启动
5  docker compose up -d
6
7  # 查看状态
8  docker compose ps

```

```
1 # 查看后端日志
2 docker compose logs -f obe-backend
3
4 # 查看宿主机后端日志
5 tail -f /works/course-oss/logs/backend/backend.log
6
7 # 停止,会停止容器, 慎用
8 docker compose down
9
10 # 停止并清除数据, 慎用
11 docker compose down -v
```

访问地址:

<http://10.65.199.3:5173/>

3.8 运维

3.8.1 更新后端

执行以下命令stop后端:

```
1 # 进入项目目录
2 cd /works/course-oss/
3 # 停止后端
4 docker compose stop obe-backend
```

```
^Croot@iZbp12sl915kbpstfajg3Z:/works/course-oss# docker compose stop obe-backend
[+] stop 1/1
✓ Container obe-backend Stopped
root@iZbp12sl915kbpstfajg3Z:/works/course-oss#
```

打包时, 选择prod:

```
application-prod.yml x docker-compose.yml x application-dev.yml x application.yml x
1 server:
2   port: 8080
3
4 spring:
5   profiles:
6     active: prod
7   jackson:
8     default-property-inclusion: non_null
9     date-format: yyyy-MM-dd HH:mm:ss
10    time-zone: Asia/Shanghai
```

更换后端jar包：

47.96.66.16					
/works/course-oss/logs/backend					
名称	大小	类型	修改时间	属性	所有者
obe-platform-1.0.0.jar	64.07MB	Executabl...	2026-05-24, 下午 9:32	-rw-r--r--	root

重新构建后端镜像：

```
Bash
1 # 重新构建后端镜像（仅构建后端，即只执行docker-compose.yml中与后端有关的命令，不影响其他服务）
2 docker compose build obe-backend
```

```
root@iZbp12s1915kbpstfajg3Z:/works/course-oss# docker compose build obe-backend
[+] Building 11.7s (10/10) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 528B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 289B 0.0s
=> [internal] load metadata for docker.io/library/openjdk:17.0.2 0.1s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/3] FROM docker.io/library/openjdk:17.0.2@sha256:528707081fdb9562eb819128a9f85ae7fe000e2fbaeaf9f87662e7b3f38cb7d8 0.2s
=> => resolve docker.io/library/openjdk:17.0.2@sha256:528707081fdb9562eb819128a9f85ae7fe000e2fbaeaf9f87662e7b3f38cb7d8 0.2s
=> [internal] load build context 3.4s
=> => transferring context: 67.19MB 3.3s
=> CACHED [2/3] RUN ln -snf /usr/share/zoneinfo/Asia/Shanghai /etc/localtime && echo Asia/Shanghai > /etc/timezone 0.0s
=> [3/3] COPY obe-platform-1.0.0.jar /app.jar 0.6s
=> exporting to image 6.1s
=> => exporting layers 5.0s
=> => exporting manifest sha256:b7507b83248d0b0c4b1684549d8724af62509bce40c55ed53aa66dec865fa455 0.0s
=> => exporting config sha256:8bf2f47c1950737ef37ef89f753c9201d56d091545267c40b4d26fc5046c9ab0 0.0s
=> => exporting attestation manifest sha256:e606db49600b3b6cf8e903f6c3624ff1e9ee5f6612ca084395f371ddb9d3252 0.0s
=> => exporting manifest list sha256:c1b5e6cc79815dbf77bdf15a0c849260535f6fe69e4e5dcf35d78a778a7edaee 0.0s
=> => naming to docker.io/library/course-oss-obe-backend:latest 0.0s
=> => unpacking to docker.io/library/course-oss-obe-backend:latest 0.9s
=> resolving provenance for metadata file 0.2s
[+] build 1/1 11.9s
Image course-oss-obe-backend Built
```

启动更新后的后端容器：

```
1 docker compose up -d obe-backend
```

docker跟踪后端日志：

```
1 # 查看后端日志
2 docker logs -f obe-backend
```

3.8.2 前端

我们在docker-compose.yml中，配置了宿主机目录映射容器目录：

```
volumes:
  - "/works/course-oss/nginx/nginx.conf:/etc/nginx/nginx.conf"
  - "/works/course-oss/nginx/logs:/var/log/nginx"
  - "/works/course-oss/frontend:/usr/share/nginx/html/frontend" # 挂载Vue静态文件
depends_on:
  - obe-backend
```

因此更新前端和传统部署方式相同：

```
1 # 进入前端目录
2 cd /works/course-oss/frontend
3
4 # 更换打包的文件（dist里面的所有文件）
5
6 # 重启nginx
7 docker restart obe-nginx
```

四、application-prod.yml 配置


```
1  spring:
2    datasource:
3      url: jdbc:mysql://obe-mysql:3306/obe_platform?useSSL=false&allowPublic
      KeyRetrieval=true&serverTimezone=Asia/Shanghai
4      username: root
5      password: Qzb123456*
6      driver-class-name: com.mysql.cj.jdbc.Driver
7      hikari:
8        maximum-pool-size: 20
9        minimum-idle: 10
10
11  #mybatis-plus:
12  #  configuration:
13  #    log-impl: org.apache.ibatis.logging.noLogging.NoLoggingImpl
14
15  logging:
16    level:
17      com.obe.platform: debug
18    file:
19      name: /app/logs/backend.log
20
```

- 传统部署: `MYSQL_HOST` 默认 `localhost`
- Docker 部署: `docker-compose` 注入 `MYSQL_HOST=obe-mysql` (容器名)

五、两种方式对比

维度	传统部署	Docker 部署
后端管理	<code>systemctl start obe-platform</code>	<code>docker compose up -d</code>
日志查看	<code>tail -f logs/backend d/backend.log</code>	<code>docker compose logs -f</code>
数据库	系统安装 MySQL	容器内 MySQL (数据挂载宿主机)

网络	本地回环	<code>obe-net</code> 桥接网络
适用场景	生产环境长期运行	快速部署 / 开发测试 / 演示